

Computational Wind Engineering

Fluid flow governing equations

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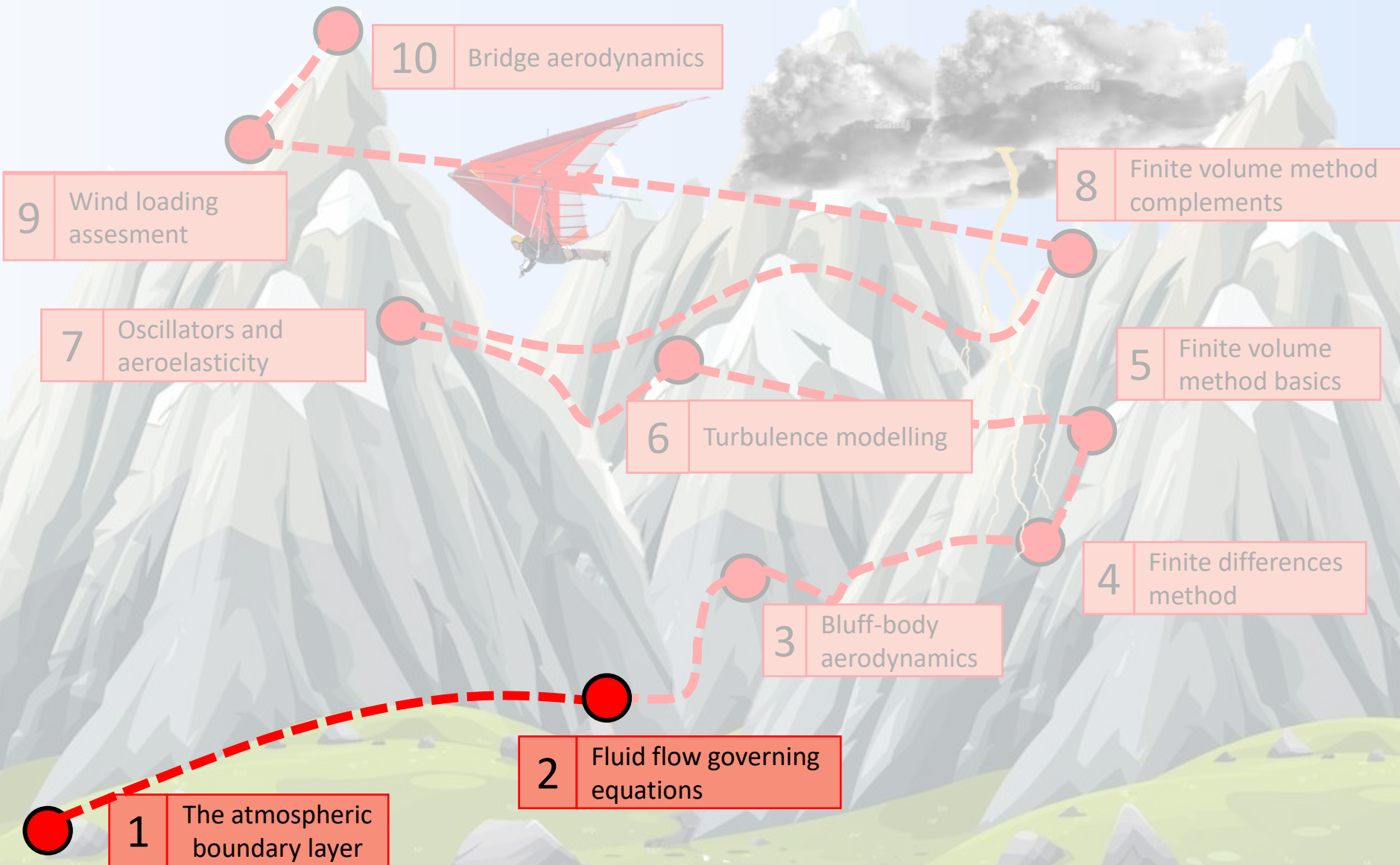
- Balance equations
- Mass and momentum balance
- Molecular viscosity
- Navier-Stokes equations
- Turbulence

Map of the course

Wind Engineering

Fluid Dynamics

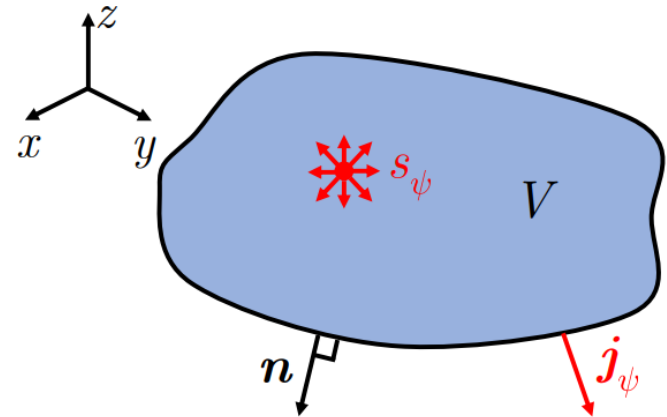
Numerical methods



Let us consider a **volume of fluid**, V , fixed in space and bounded by the surface S . Let us assume a scalar quantity ψ is defined over the volume, which represents a density per unit volume, e.g. ink concentration per unit volume.

The total amount of ψ (e.g. ink) within the volume is

$$\int_V \psi dV.$$



Additionally, some ψ might be **flowing through the boundaries**. We indicate as $\mathbf{j}_\psi$ the flux of ψ per unit area. The total amount of ψ exiting the volume in unit time is

$$\int_S \mathbf{j}_\psi^T \mathbf{n} dS,$$

Additionally, some amount of ψ might be **produced within the control volume**. Denoting as s_ψ the production per unit volume per unit time, the total amount of produced ψ is

$$\int_V s_\psi dV,$$

It is thus possible to write the balance of ψ within the control volume as

$$\frac{d}{dt} \int_V \psi dV = - \int_S \mathbf{j}_\psi^T \mathbf{n} dS + \int_V s_\psi dV,$$

This equation simply states that the variation of ψ within the control volume is the integral of the production minus the amount which was lost through the boundary. **It is the balance equation for ψ in integral form.**

For the following developments, it is useful to **split the fluxes through the surfaces into two contributions**

$$\dot{\mathbf{j}}_{\psi} = \dot{\mathbf{j}}_{\psi,a} + \dot{\mathbf{j}}_{\psi,na},$$

where $\dot{\mathbf{j}}_{\psi,a}$ is the advective flux and $\dot{\mathbf{j}}_{\psi,na}$ is the non-advective flux. Advective fluxes represent the amount of ψ exiting the volume due to the fluid motion through the control volume surface and reads

$$\dot{\mathbf{j}}_{\psi,a} = \psi \mathbf{u},$$

with $\mathbf{u} = [u, v, w]^T$ velocity vector. Finally, it is common to express

$$S_{\psi} = S_{\psi,p} - S_{\psi,d}$$

being the first one a **production term** and the second one a **destruction term**.

The balance equation can be thus rewritten as

$$\frac{d}{dt} \int_V \psi dV + \int_S \psi \mathbf{u}^T \mathbf{n} dS + \int_S \mathbf{j}_{\psi, na}^T \mathbf{n} dS = \int_V s_{\psi, p} - s_{\psi, d} dV,$$

We now introduce (without demonstration), the **Divergence Theorem**, a particular case of the Gauss-Green Theorem. For a general vectorial quantity $\boldsymbol{\gamma}$, under some regularity conditions

$$\int_S \boldsymbol{\gamma}^T \mathbf{n} dS = \int_V \boldsymbol{\nabla} \cdot \boldsymbol{\gamma} dV, \quad \text{with} \quad \boldsymbol{\nabla} = \left[\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right]^T,$$

so that, developing the products

$$\boldsymbol{\nabla} \cdot \boldsymbol{\gamma} = \begin{bmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{bmatrix} \begin{bmatrix} \gamma_x \\ \gamma_y \\ \gamma_z \end{bmatrix} = \frac{\partial \gamma_x}{\partial x} + \frac{\partial \gamma_y}{\partial y} + \frac{\partial \gamma_z}{\partial z},$$

We can thus write

$$\frac{d}{dt} \int_V \psi dV + \int_S \psi \mathbf{u}^T \mathbf{n} dS + \int_S \mathbf{j}_{\psi, na}^T \mathbf{n} dS = \int_V s_{\psi, p} - s_{\psi, d} dV,$$

which leads to

$$\frac{d}{dt} \int_V \psi dV + \int_V \nabla \cdot (\psi \mathbf{u}) dV + \int_V \nabla \cdot \mathbf{j}_{\psi, na} dV = \int_V s_{\psi, p} - s_{\psi, d} dV.$$

Taking the time derivative inside the integral, as the equation must hold for every control volume, we obtain

$$\frac{\partial \psi}{\partial t} + \nabla \cdot (\psi \mathbf{u}) + \nabla \cdot \mathbf{j}_{\psi, na} = s_{\psi, p} - s_{\psi, d},$$

which represents the balance equation for ψ in **differential form**.

The first quantity we want to write a balance for is **mass**. The balance is obtained substituting $\psi = \rho$ with ρ fluid density. We thus obtain

$$\cancel{\frac{\partial \rho}{\partial t}} + \nabla \cdot (\rho \mathbf{u}) + \nabla \cdot \cancel{\mathbf{j}_{\rho,na}} = \cancel{s_{\rho,p}} - \cancel{s_{\rho,d}}.$$

However, we can usually assume null mass source terms and, for Mach number Ma (ratio between fluid speed and speed of sound), <0.3 , the fluid has constant density (incompressibility) with good approximation.

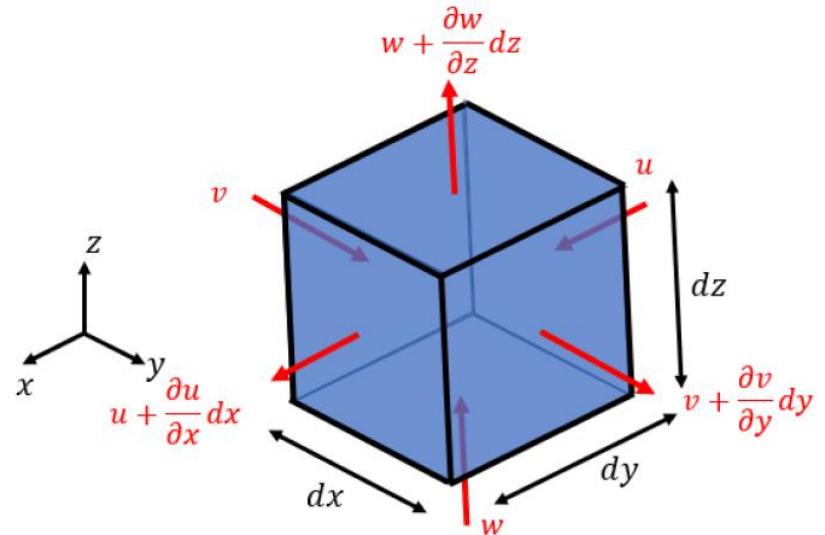
Further, $\mathbf{j}_{\psi,na}$ is null for mass transport, so finally

$$\nabla \cdot \mathbf{u} = 0,$$

which is known as **divergence-free condition**.

It is possible to obtain the divergence-free condition in a more intuitive way. To this purpose, let us consider the control volume reported in the figure.

Assuming the fluid to be incompressible, **the sum of all fluxes through the control volume must be null.**



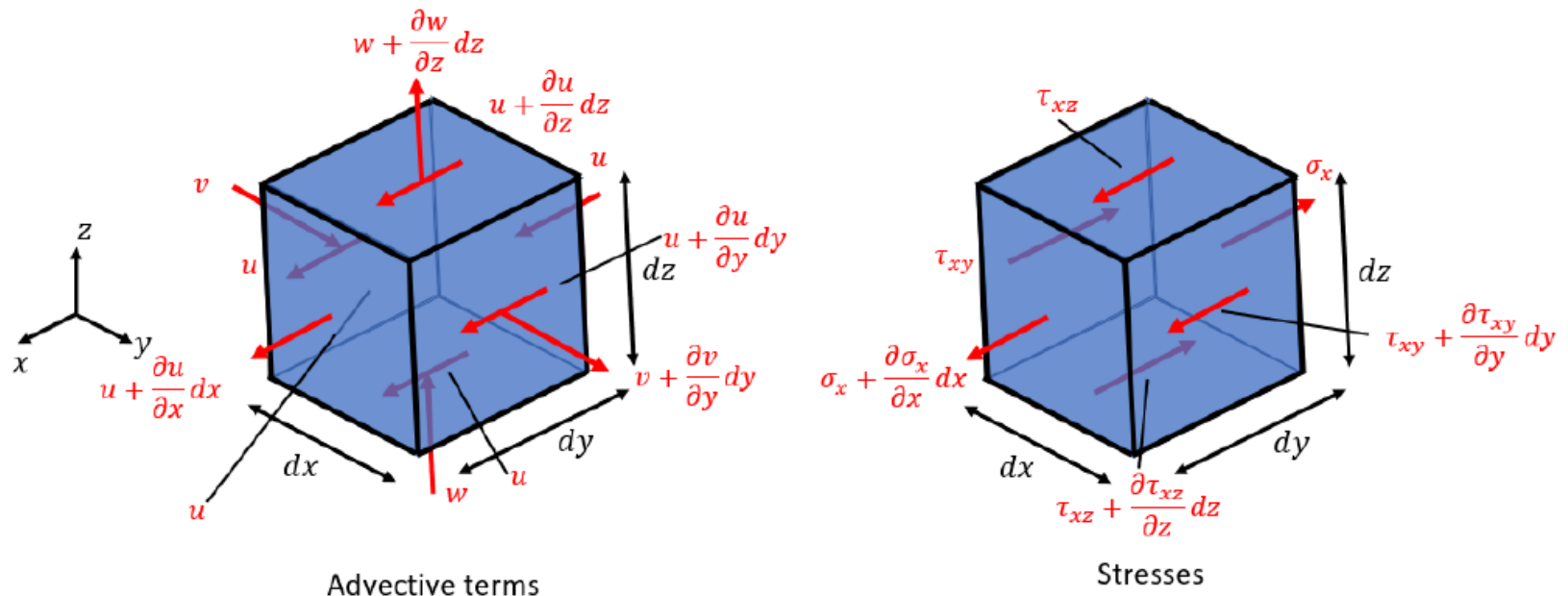
$$\left(\cancel{u + \frac{\partial u}{\partial x} dx} - \cancel{u} \right) dydz + \left(\cancel{v + \frac{\partial v}{\partial y} dy} - \cancel{v} \right) dx dz + \left(\cancel{w + \frac{\partial w}{\partial z} dz} - \cancel{w} \right) dx dy = 0,$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0.$$

Momentum is defined as $\rho \mathbf{u}$, i.e. it is a vectorial quantity. We can write the balance for the first component of the momentum choosing $\psi = \rho u$.

$$\frac{\partial(\rho u)}{\partial t} + \nabla \cdot (\rho u \mathbf{u}) + \nabla \cdot \mathbf{j}_{\rho u, na} = s_{\rho u, p} - s_{\rho u, d}.$$

However, we here directly proceed in a more intuitive manner. The x-momentum fluxes over an infinitesimal control volume are



The variation within the infinitesimal control volume is

$$\rho \frac{\partial u}{\partial t} dx dy dz.$$

The advective fluxes through faces with normal directed along x are

$$\left[\left(u + \frac{\partial u}{\partial x} dx \right)^2 - u^2 \right] \rho dy dz.$$

which expanding the square leads to

$$\left[\cancel{u^2} + 2u \frac{\partial u}{\partial x} dx + \cancel{\left(\frac{\partial u}{\partial x} dx \right)^2} - \cancel{u^2} \right] \rho dy dz = 2u \frac{\partial u}{\partial x} \rho dx dy dz.$$

Analogously, those through faces with normal directed along y are

$$\left[\left(u + \frac{\partial u}{\partial y} dy \right) \left(v + \frac{\partial v}{\partial y} dy \right) - uv \right] \rho dx dz. \quad \rightarrow \quad \left(u \frac{\partial v}{\partial y} + v \frac{\partial u}{\partial y} \right) \rho dx dy dz.$$

The form taken by the advective fluxes across faces with normal directed along z are similar to those already found for y , so that, summing all together

$$\left[2u \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} u + \frac{\partial u}{\partial x} v + \frac{\partial w}{\partial z} u + \frac{\partial u}{\partial x} w \right] \rho dx dy dz = \left[u \frac{\partial u}{\partial x} + \frac{\partial u}{\partial x} v + \frac{\partial u}{\partial x} w \right] \rho dx dy dz,$$

in which the divergence-free condition has been adopted.
Considering now non advective fluxes, i.e. **stresses**, we have

$$- \left[\sigma_x + \frac{\partial \sigma_x}{\partial x} dx - \sigma_x \right] dy dz + \left[\tau_{xy} + \frac{\partial \tau_{xy}}{\partial y} dy - \tau_{xy} \right] dx dz + \left[\tau_{xz} + \frac{\partial \tau_{xz}}{\partial z} dz - \tau_{xz} \right] dx dy,$$

and, finally, volume forces are

$$f_x dx dy dz.$$

Summing everything, we obtain

$$\rho \frac{\partial u}{\partial t} + \rho \frac{\partial u}{\partial x} u + \rho \frac{\partial u}{\partial y} v + \rho \frac{\partial u}{\partial z} w - \frac{\partial \sigma_x}{\partial x} - \frac{\partial \tau_{xy}}{\partial y} - \frac{\partial \tau_{xz}}{\partial z} = f_x$$

Considering now all components we can write

$$\begin{cases} \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} u + \frac{\partial u}{\partial y} v + \frac{\partial u}{\partial z} w - \frac{1}{\rho} \left[\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} \right] = \frac{f_x}{\rho} \\ \frac{\partial v}{\partial t} + \frac{\partial v}{\partial x} u + \frac{\partial v}{\partial y} v + \frac{\partial v}{\partial z} w - \frac{1}{\rho} \left[\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} \right] = \frac{f_y}{\rho} \\ \frac{\partial w}{\partial t} + \frac{\partial w}{\partial x} u + \frac{\partial w}{\partial y} v + \frac{\partial w}{\partial z} w - \frac{1}{\rho} \left[\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_z}{\partial z} \right] = \frac{f_z}{\rho} \end{cases}$$

The equation can be written using matrix notation as

$$\frac{\partial}{\partial t} \begin{pmatrix} u \\ v \\ w \end{pmatrix} + \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} - \frac{1}{\rho} \begin{bmatrix} \frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} \\ \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} \\ \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_z}{\partial z} \end{bmatrix} = \frac{1}{\rho} \begin{bmatrix} f_x \\ f_y \\ f_z \end{bmatrix}$$

Or simply as

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla(\mathbf{u}) \cdot \mathbf{u} - \frac{1}{\rho} \nabla \cdot \boldsymbol{\sigma} = \frac{\mathbf{f}}{\rho}.$$

Notice that in our convention $\nabla(\mathbf{u})$ represents the gradient of $\mathbf{u}$.

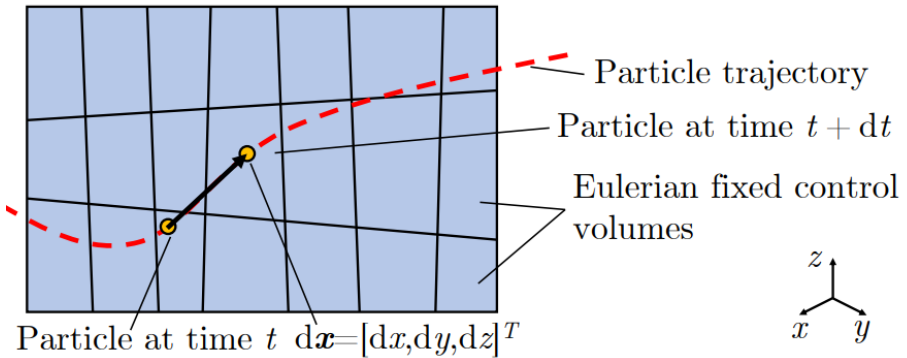
$$\frac{\partial}{\partial t} \begin{pmatrix} u \\ v \\ w \end{pmatrix} + \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{bmatrix} \begin{pmatrix} u \\ v \\ w \end{pmatrix} - \frac{1}{\rho} \begin{bmatrix} \frac{\partial \sigma_x}{\partial x} & \frac{\partial \tau_{xy}}{\partial y} & \frac{\partial \tau_{xz}}{\partial z} \\ \frac{\partial \tau_{xy}}{\partial x} & \frac{\partial \sigma_y}{\partial y} & \frac{\partial \tau_{yz}}{\partial z} \\ \frac{\partial \tau_{xz}}{\partial x} & \frac{\partial \tau_{yz}}{\partial y} & \frac{\partial \sigma_z}{\partial z} \end{bmatrix} = \frac{1}{\rho} \begin{pmatrix} f_x \\ f_y \\ f_z \end{pmatrix},$$

Notice that momentum and mass balance are not enough to solve for the fluid motion, as we have more unknowns than equations

$$\begin{cases} \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} u + \frac{\partial u}{\partial y} v + \frac{\partial u}{\partial z} w - \frac{1}{\rho} \left[\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} \right] = \frac{f_x}{\rho} \\ \frac{\partial v}{\partial t} + \frac{\partial v}{\partial x} u + \frac{\partial v}{\partial y} v + \frac{\partial v}{\partial z} w - \frac{1}{\rho} \left[\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} \right] = \frac{f_y}{\rho} \\ \frac{\partial w}{\partial t} + \frac{\partial w}{\partial x} u + \frac{\partial w}{\partial y} v + \frac{\partial w}{\partial z} w - \frac{1}{\rho} \left[\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_z}{\partial z} \right] = \frac{f_z}{\rho} \end{cases}$$

To close the system we need to select a fluid model.

It is finally useful to provide a different point of view regarding the advective terms. Let us start from Newton second law of dynamics



$$m\mathbf{a} = \mathbf{r}, \quad \longrightarrow \quad m \frac{D\mathbf{u}}{Dt} = \mathbf{r},$$

$$\frac{Du}{Dt} = \frac{u(\mathbf{x}(t + dt), t + dt) - u(\mathbf{x}(t), t)}{dt} \quad \text{with} \quad \mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{u}dt.$$

$$\frac{Du}{Dt} = \frac{u(x + udt, y + vdt, z + wdt, t + dt) - u(x, y, z, t)}{dt},$$

$$\frac{Du}{Dt} = \frac{u(x, y, z, t) + \frac{\partial u}{\partial x}udt + \frac{\partial u}{\partial y}vdt + \frac{\partial u}{\partial z}wdt + \frac{\partial u}{\partial t}dt - u(x, y, z, t)}{dt},$$

$$\frac{Du}{Dt} = \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x}u + \frac{\partial u}{\partial y}v + \frac{\partial u}{\partial z}w = \frac{\partial u}{\partial t} + \nabla(u) \cdot \mathbf{u}.$$

For incompressible fluids, the sum of time variation and advective terms is the total derivative.

The simplest possible fluid model is obtained assuming

$$\underline{\sigma} = -p\underline{I},$$

with such a choice

$$\nabla \cdot \underline{\sigma} = -\nabla p.$$

which substituted into the momentum balance equation leads to

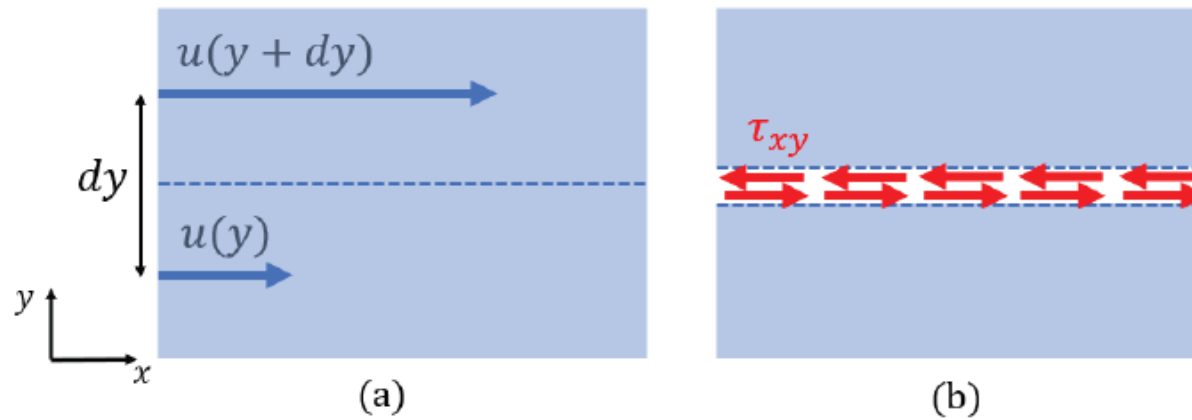
$$\frac{\partial \underline{u}}{\partial t} + \nabla(\underline{u}) \underline{u} + \frac{1}{\rho} \nabla p = \frac{\underline{f}}{\rho},$$

In components, the governing equations become

$$\left\{ \begin{array}{l} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \\ \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x}u + \frac{\partial u}{\partial y}v + \frac{\partial u}{\partial z}w + \frac{1}{\rho} \frac{\partial p}{\partial x} = \frac{f_x}{\rho} \\ \frac{\partial v}{\partial t} + \frac{\partial v}{\partial x}u + \frac{\partial v}{\partial y}v + \frac{\partial v}{\partial z}w + \frac{1}{\rho} \frac{\partial p}{\partial y} = \frac{f_y}{\rho} \\ \frac{\partial w}{\partial t} + \frac{\partial w}{\partial x}u + \frac{\partial w}{\partial y}v + \frac{\partial w}{\partial z}w + \frac{1}{\rho} \frac{\partial p}{\partial z} = \frac{f_z}{\rho} \end{array} \right.$$

What happens if
the fluid is at rest?

Euler equations neglect a very important contribution to the non-advective momentum fluxes. **Adjacent fluid fillets “friction” one against the other, i.e. the fluid is viscous.**

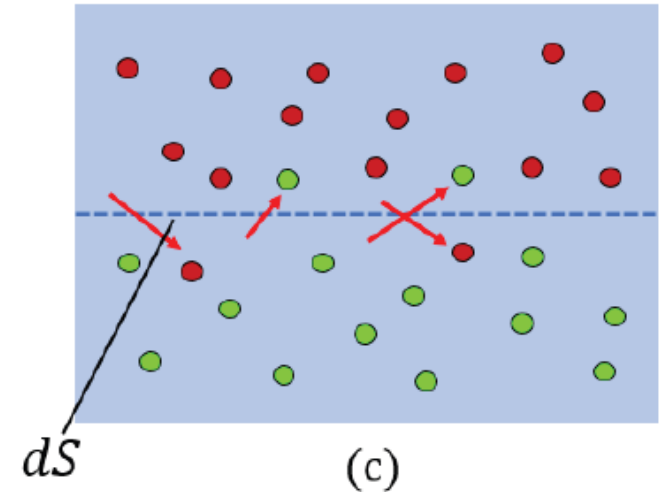


and, according to Newton's law, viscous stresses are

$$\tau_{xy} = \mu \frac{du}{dy}, \quad \text{or alternatively} \quad \tau_{xy} = \nu \rho \frac{du}{dy},$$

being μ and ν the (dynamic) viscosity and the kinematic viscosity parameters.

We now seek for a micro-mechanical explanation of this result. Let us consider a fluid flow as the motion of a multitude of fluid particles.



1. molecules are characterized by the velocity typical of the fluid fillet they come from;
2. they typically move at the thermal speed $|\mathbf{u}_{th}|$, relative to the fluid fillet overall motion;
3. the direction of the thermal motion is random and equally probable in all directions;
4. typically, molecules travel a distance λ , the free-mean path, before colliding with another particle, so equalizing its speed with the surrounding fluid particles.

Under these assumptions, it is possible to demonstrate that the momentum flux between two adjacent fluid fillets is

$$\tau_{xy} = \mu \frac{dU}{dy} \approx \frac{|\mathbf{u}_{th}|}{2} \rho \frac{dU}{dy} \lambda \quad \Rightarrow \quad \mu = \frac{1}{2} \rho |\mathbf{u}_{th}| \lambda.$$

The extension of the Newton's law for viscous stresses (which is a constitutive law, i.e. like Hook's law for elasticity) to the three dimensional case for incompressible fluids is

$$\underline{\sigma}_v = \mu \left(\nabla(\mathbf{u}) + \nabla(\mathbf{u})^T \right) = 2\mu \underline{\nabla}_s(\mathbf{u}),$$

where $\underline{\nabla}_s(\mathbf{u})$ is the symmetric part of the gradient of $\mathbf{u}$.

$$\underline{\nabla}_s(\mathbf{u}) = \frac{1}{2} \left[\nabla(\mathbf{u}) + \nabla(\mathbf{u})^T \right].$$

Stresses can be thus written as

$$\underline{\sigma} = -p\underline{\mathbf{I}} + 2\mu \underline{\nabla}_s(\mathbf{u})$$

leading to

$$\begin{cases} \nabla \cdot \mathbf{u} = 0 \\ \frac{\partial \mathbf{u}}{\partial t} + \nabla(\mathbf{u})\mathbf{u} - \frac{1}{\rho} \nabla \cdot [-p\underline{\mathbf{I}} + 2\mu \underline{\nabla}_s(\mathbf{u})] = \frac{\mathbf{f}}{\rho}, \end{cases}$$

Recalling the result obtained for Euler's model, we can also write

$$\begin{cases} \nabla \cdot \mathbf{u} = 0 \\ \frac{\partial \mathbf{u}}{\partial t} + \nabla(\mathbf{u})\mathbf{u} + \frac{\nabla(p)}{\rho} - \frac{1}{\rho} \nabla \cdot [2\mu \nabla_s(\mathbf{u})] = \frac{\mathbf{f}}{\rho} \end{cases}$$

which, written in components and developing the derivatives, reads

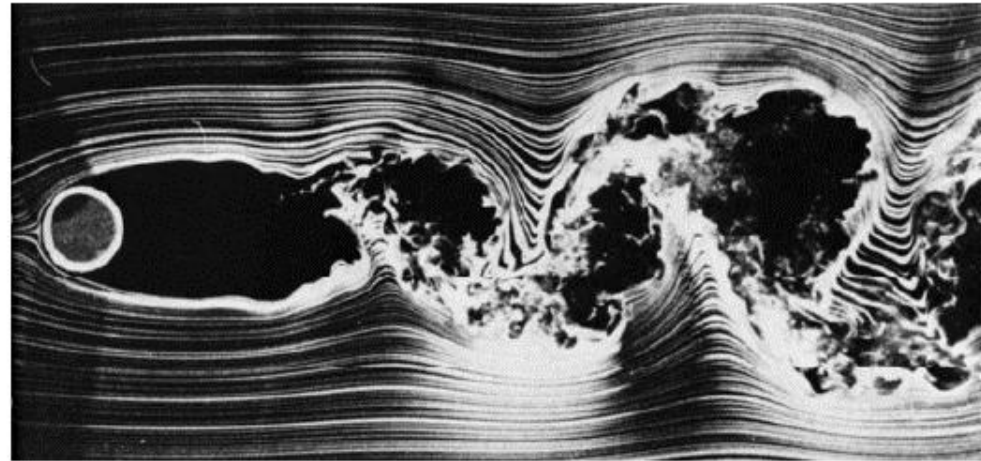
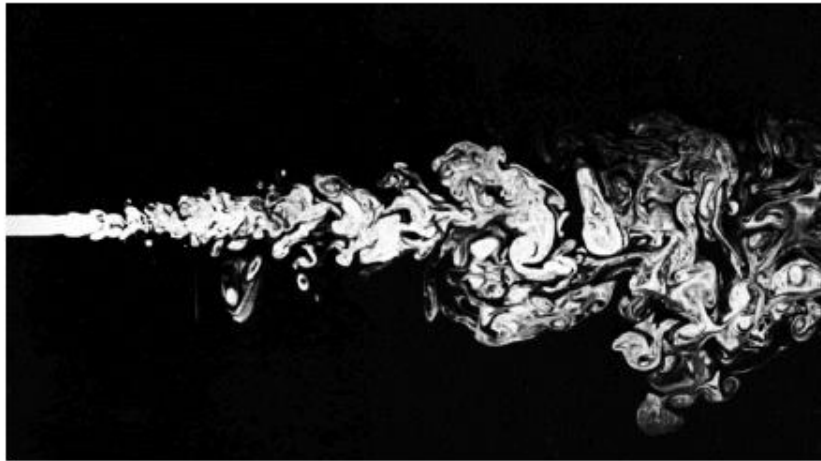
$$\begin{cases} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \\ \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x}u + \frac{\partial u}{\partial y}v + \frac{\partial u}{\partial z}w + \frac{1}{\rho} \frac{\partial p}{\partial x} - \frac{\mu}{\rho} \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) = \frac{f_x}{\rho} \\ \frac{\partial v}{\partial t} + \frac{\partial v}{\partial x}u + \frac{\partial v}{\partial y}v + \frac{\partial v}{\partial z}w + \frac{1}{\rho} \frac{\partial p}{\partial y} - \frac{\mu}{\rho} \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) = \frac{f_y}{\rho} \\ \frac{\partial w}{\partial t} + \frac{\partial w}{\partial x}u + \frac{\partial w}{\partial y}v + \frac{\partial w}{\partial z}w + \frac{1}{\rho} \frac{\partial p}{\partial z} - \frac{\mu}{\rho} \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) = \frac{f_z}{\rho}. \end{cases}$$

or, more concisely

$$\begin{cases} \nabla \cdot \mathbf{u} = 0 \\ \frac{\partial \mathbf{u}}{\partial t} + \nabla(\mathbf{u})\mathbf{u} + \frac{\nabla p}{\rho} - \nu \nabla^2(\mathbf{u}) = \frac{\mathbf{f}}{\rho}, \end{cases}$$

being ∇^2 the Laplace operator. **These equations are known as Navier-Stokes equations.**

It is notoriously extremely hard to obtain analytical results from the Navier-Stokes equations. We can, however, study **some emergent properties of the solution which, based on the observation of real flows, appear to develop systematically.**



In particular, under very common circumstances, the flow develops fluctuations of the velocity and pressure fields due to the **presence of structures which might be described as eddies.**

Left image photographed by Dimotakis P., Lye R.C and Papantoniou D.Z. 1981. Right image photographed by Corke, T. and Nagib, H. Both images are collected in An Album of Fluid Motion (Van Dyke 1982) and are used here with permission from the authors.

Turbulence is always characterized by:

1. **unsteadiness** - turbulent fluctuations are never steady, nor they are periodic;
2. **three-dimensionality** - turbulence is always three-dimensional;
3. **randomness** - the Navier-Stokes system is deterministic, but the developed fluctuations appear to be random;
4. **chaoticity** - the flow development strongly depends on the initial and boundary conditions;
5. **fractality** - it is not possible to understand how big a picture of a turbulent flow is just by looking at it. Patterns repeat identically at every scale (i.e. it is multi-scale and self-similar).

Due to the aforementioned properties, **turbulence must be described on statistical bases**. In particular, we write

$$\mathbf{u}(\mathbf{x}, t) = \mathbf{U}(\mathbf{x}) + \mathbf{u}'(\mathbf{x}, t),$$

and define the **Turbulent Kinetic Energy** (per unit volume and density) as

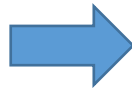
$$k = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_0^T \mathbf{u}'^T \mathbf{u}' dt = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_0^T (u'^2 + v'^2 + w'^2) dt,$$

whose dimensions are $\left(\frac{m}{s}\right)^2$. **We can also define ε , the rate of dissipation of k** , which thus has dimensions

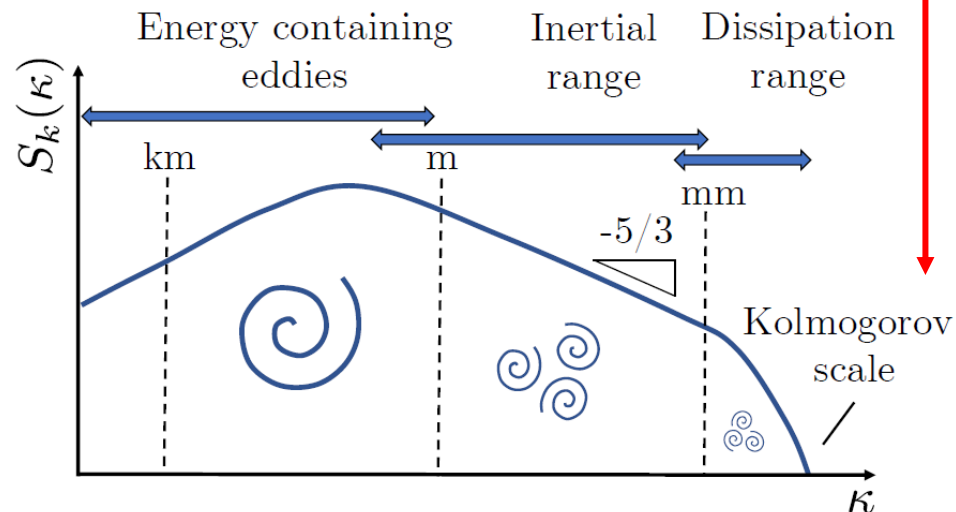
$$\frac{m^2}{s^3}$$

The turbulent kinetic energy is fed by the mean flow and the energy cascades to smaller scales up to dissipation into heat. Kolmogorov understood that the process must be governed by only two parameters ε and ν . From the dimensional point of view

$$\varepsilon [L^2 T^{-3}]$$
$$\nu [L^2 T^{-1}]$$



$$l_K = \left(\frac{\nu^3}{\varepsilon} \right)^{1/4}$$



It is finally possible to notice that **the presence of turbulence can be assimilated to molecular random motion, which we already saw is the responsible for momentum diffusion, i.e. viscosity.**

With such an observation, Boussinesq argued that the presence of turbulence might be represented as an additional viscosity, i.e. the **turbulent viscosity**.

On dimensional ground and in analogy to the molecular case

$$\mu_t = C_\mu \rho l_t u_t,$$

where C_μ is a constant to be calibrated, while l_t (**mixing length**) and u_t are some sort of turbulence length and velocity, reminiscent of the free mean path and the thermal velocity. Finally, the effective viscosity is

$$\mu_e = \mu + \mu_t,$$